

Can grapefruit juice influence ethinylestradiol bioavailability?

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The effects of grapefruit juice on the bioavailability of 17 alpha-ethinylestradiol (EE2) after a single oral administration of 50 micrograms EE2 have been investigated. The pharmacokinetics of EE2 were studied in an open, randomized, cross-over study in which 13 healthy volunteers were administered the drug with herbal tea or grapefruit juice (naringin, 887 mg/ml). In contrast to herbal tea, grapefruit juice increased the peak plasma concentration (C_{max}) significantly to 137% (mean; range 64% to 214%, $p = 0.0088$) and increased the area under plasma concentration-time curve from 0 to 8 hours (AUC₀₋₈) to 128% (mean; range 81% to 180%, $p = 0.0186$). This study shows that grapefruit juice increases the bioavailable amount of EE2. A possible explanation may be that grapefruit juice inhibits the metabolic degradation of EE2. Whether the increased bioavailability of EE2 following grapefruit juice administration is of clinical importance should be investigated in long-term studies.